

Md. Abul Hasan

Senior Staff Software Engineer

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Portfolio & case studies: ahasan09.github.io · Dhaka, Bangladesh



13+

Years shipping production software

1M+

Users on the public Arkivportalen portal I own

100K+

EU users on the Swiss Life Select platform I led

5

Codebases modernized — 4x Angular 14→21, 1x latest React

I architect and ship frontends for nation-scale platforms, and work across the stack when delivery requires it. At senior staff level I focus on architecture that survives scale, delivery pressure, and team changes — writing production code, driving technical direction, and mentoring engineers. I run an agentic development workflow end-to-end: an agent pipeline picks up a Jira ticket, analyzes the requirement, generates a design spec and plan, implements it, then runs code review and security review, while I direct the architecture and own every merge. Every role in my career has run on Agile/Scrum — sprints, retros, and regular releases.

EXPERIENCE

Senior Staff Software Engineer · Cefalo

Aug 2022 – Present

Client: Norwegian National Archive (Asta), Norway — nation-scale archive platform, 5 modules, microservice-backed

- Own the frontend of all 5 modules, plus the Arkivportalen portal and requisition backends end-to-end — leading the team day-to-day, with oversight of backend work on the remaining modules
- Drove Angular 14→21 and React migrations across all 5 repositories — pilot repo first, then the rest in overlap; ~2 months with a team cut from 10 to 5, keeping Angular Material on v14 as a deliberate trade-off against MDC breaking changes
- Built a Kafka CDC pipeline (Debezium, Kafka Connect) keeping Elasticsearch and cross-module shared data in sync with PostgreSQL/MongoDB
- Built a digital-archive middleware in Python — pulls portal data into its own PostgreSQL and reconciles publish status against an external digital-media system, with per-item checkpointing and retries; the portal never calls the external system
- Built the dynamic permission engine and Elasticsearch-backed search; implemented WebSocket live notifications; Google Analytics integration and UI improvements on the portal
- Set up self-hosted Bitbucket Pipelines runners on Azure VMs; standardized testing and CI across repos — Playwright E2E, SonarQube, SBOM checks, shared lint/test setup
- Run an agentic delivery workflow (Claude Code + Jira MCP): ticket → analysis → design spec → plan → implementation → code + security review — every stage a custom skill I built; Figma MCP for design-to-code; adopted by the whole team
- Outcome: production system serving 10,000+ archivists and a public portal with 1M+ users; migration roadmap delivered with a 50% leaner team

Stack: Angular, React, TypeScript, Kotlin, Java 21, Spring Boot, Python, PostgreSQL, MongoDB, Elasticsearch, Kafka, ActiveMQ

Principal Software Engineer · SELISE Digital Platforms

2018 – Jul 2022

Client: Swiss Life Select fintech platform, EU — multi-tenant, microservice-backed advisor & customer portal

- Owned core frontend architecture for a platform serving 100,000+ users across the EU on 4 tenant builds; led the 12-member frontend team within a ~20-member product team
- Built the company-wide Angular frontend architecture (SSR-capable) used by all SELISE teams — teams published shared components, and common packages (auth, SSO, SignalR-integrated notifications, dynamic datatable, dialog) shipped on an internal npm registry
- Delivered SSO (OIDC), multi-tenancy via a shared component library with separate per-tenant builds, PWA, and push notifications
- Real-time delivery via SignalR + RabbitMQ; data-intensive dashboards with Chart.js and AG Grid
- Migrated 4 projects from Angular 2 to 12

Stack: Angular, TypeScript, .NET Core, C#, MongoDB, Redis, RabbitMQ, SignalR, JWT, PWA

- **Senior Software Engineer** — Swiss Life ePrivateWealth (2017): internal/external insurance portal; AngularJS + .NET Core microservices, SignalR real-time
- **Software Engineer** — Ruf Informatik E-Service (2016): reservation platform, social-feed aggregation, public statistics dashboards
- **Developer** — Computer Source Bangladesh (2013–2015): end-to-end supply-chain ERP — inventory, RMA, sales, POS, accounts, PMS

ARCHITECTURE CASE STUDIES

- **Migrating 5 codebases to Angular 21 with half the team** — sequencing, risk, CI standardization
ahasan09.github.io/case-studies/angular-migration.html
- **An agentic workflow: from Jira ticket to reviewed PR** — agent pipeline with a human owning every merge
ahasan09.github.io/case-studies/agentic-workflow.html
- **CDC with Kafka: keeping Elasticsearch in sync** — why change data capture beat dual writes
ahasan09.github.io/case-studies/kafka-cdc-search.html
- **A pull-only middleware: publish status behind a rate-limited API** — per-item checkpointing at 1M-user scale
ahasan09.github.io/case-studies/archive-middleware.html
- **A multi-tenant fintech platform for 100K EU users** — SSO, tenancy isolation, PWA, real-time
ahasan09.github.io/case-studies/swiss-life-platform.html

STACK

Frontend	Angular, React, TypeScript, JavaScript, RxJS, Redux, Angular Material, PrimeNG, Chakra UI, Bootstrap, PWA, SSR
Backend	NestJS, Node.js, Express, Kotlin, Java, Spring Boot, .NET Core, C#, Python, Django, REST, GraphQL, JWT, SSO
Data & messaging	PostgreSQL, MongoDB, MSSQL, Redis, Elasticsearch, RabbitMQ, Apache Kafka, ActiveMQ, SignalR, WebSocket
Cloud & delivery	Docker, Bitbucket Pipelines, Microsoft Azure, AWS, Azure DevOps, Jenkins, GitHub Actions, Application Insights, Azure Blob Storage
Quality & practice	Playwright, Jest, Jasmine, Karma, TDD, DDD, SOLID, Design Patterns, Clean Architecture, Agile/Scrum, Agentic development (Claude Code)

EDUCATION & CERTIFICATION

- M.Sc. in Computer Science & Engineering** — Jahangirnagar University
- B.Sc. in Computer Science & Engineering** — Patuakhali Science and Technology University
- C# .NET Certification** — BASIS (2013)